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Grammatical resumptive pronouns facilitate processing before they are perceived

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Some non-obligatory resumptive pronouns (RPs) are considered *intrusive* extra-grammatical devices inserted to repair ungrammaticality [1] or facilitate processing [2]. In other cases, RPs are described as *optional* elements, assumed to be part of a language's grammatical inventory [3-4]. We investigate the validity of this distinction– i.e., whether optional RPs have a distinct grammatical status– by comparing Hebrew ORCs, where RPs are "optional," to Hebrew object WHQs, where RPs are "intrusive". If optional RPs are grammatically licensed, their availability should affect predictions during online dependency resolution; if not, they should pattern like intrusive RPs and only repair difficulty post hoc. We compare the status of Hebrew RPs in object RCs and WHQs by manipulating animacy alignment, which affects processing difficulty as well as RP preference in Hebrew ORCs [5]. Our results indicate that while gapped WHQs and RCs display similar patterns of unacceptability due to animacy misalignment, RPs overcome this unacceptability only in RCs, and not in WHQs. Correspondingly, we find that in WHQs, but not RCs, readers are considerably disrupted by misaligned arguments alone– i.e., in advance of the verb and RP (cf. 6). This suggests that optional RPs in Hebrew ORCs constitute a grammatical alternative that shapes online dependency resolution, such that their availability mitigates processing difficulty even before the dependency tail is perceived.

1 Acceptability. 162 Hebrew speakers rated the acceptability of Hebrew sentences in a 2×2×4 design: *Structure* {*WH*, *RC*}, *Tail* {*RP*, *gap*} and *Animacy alignment*: *aligned*, with an animate subject and an inanimate object, *animacy match*, where both arguments are (in)animate, and *misaligned*, with an inanimate subject and an animate object (Table 1). We found that (a) speakers' preference for gaps was larger in WHQs compared to RCs; (b) animate objects and inanimate subjects lower the acceptability of object dependencies (misalignment); (c) animacy misalignment costs are larger in WHQs than RCs; and (d) animacy misalignment costs are larger with gaps than RPs (see Table 2). The resulting pattern reveals that optional RPs in RCs overcome animacy misalignment, while RPs in WHQs are consistently rated poorly and mirror animacy costs.

2 SPR. 80 Hebrew speakers read the same materials as the acceptability study in non-cumulative SPR, only targeting clauses with RPs (*Structure*×*Alignment*) (Table 1). We found (a) longer RTs at the subject in WHQs than RCs (Est 0.02, [0.00, 0.03] 95% CI); (b) longer RTs at the subject spill-over with inanimate compared to animate subjects (Est 0.022, [0.00, 0.04] 95% CI), and following an animate compared to an inanimate object (Est 0.02,[0.00, 0.03] 95% CI); and (c) slower reading of RPs in WHQs compared to RCs (Est 0.01, [0.00, 0.02] 95% CI). While the interaction term contains zero, marginal comparisons suggest that the slowdown on inanimate subjects is only reliable in WHQs (Est 0.04, [0.01, 0.06]) and not in RCs (Est 0.01, [-0.01, 0.03] 95% CI).

Discussion. Our results show a clear distinction between optional and intrusive RPs in Hebrew. Intrusive RPs in object WHQs do not overcome the unacceptability associated with animacy misalignment, and the reading of WHQs tracks animacy difficulty at the subject– mirroring gap acceptability patterns. On the other hand, optional RPs in ORCs appear to constitute a grammatical alternative, not just a last-resort repair: they ameliorate unacceptability due to animacy misalignment, and mitigate difficulty with misaligned arguments during dependency resolution, before they are perceived. These findings suggest that the hypothesized grammatical distinction between intrusive and optional RPs modulates the processing of filler-gap dependencies, as superficially similar structures yield different processing effects depending on whether they license RPs or not (cf. 7). The lack of processing difficulty in structures that license RPs suggests that difficulty with animacy misalignment in ORCs may stem from grammatical constraints on gap licensing, and cannot be solely attributed to general memory limitations or semantic preferences.

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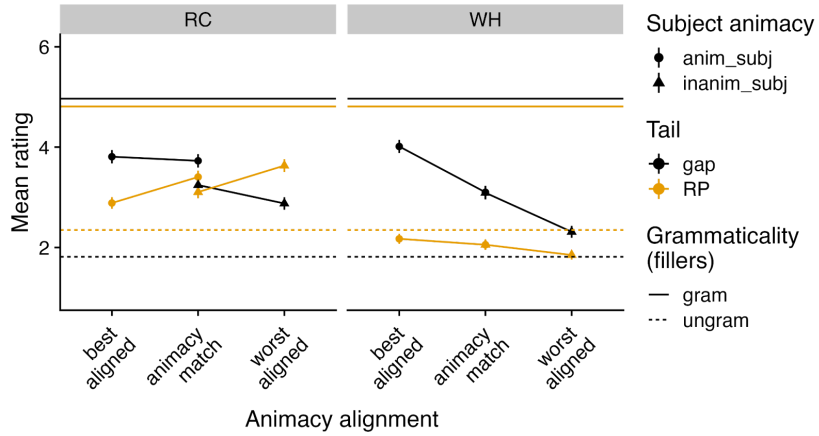
structure	alignment		dependency↓ (RP / *gap*)			
Wh-question	✓ *aligned*	I didn't see	which doll	the-teacher	hid it / ∅	from the other children
	animacy match		which doll which girl	the-car the-teacher	hid it / ∅ hid her/∅}	
	× misaligned		which girl	the-car	hid {her/∅}	
Relative clause	✓ aligned	I didn't see	the-doll	that-the-teacher	hid {it/∅}	from the other children
	animacy match		the-doll the-girl	that-the-car that-the-teacher	hid {it/∅} hid {her/∅}	
	× misaligned		the-girl	that-the-car	hid {her/∅}	

Table 1. Example stimuli, simplified, original in Hebrew. SPR study only targeted RP conditions.

Figure 1 (right). Mean acceptability ratings. Filled circles represent experimental conditions (SEM error bars). Horizontal lines plot mean ratings for fillers involving (un)grammatical gaps and RPs (IORCs which require RPs and SRCs which require gaps) as a point of reference.



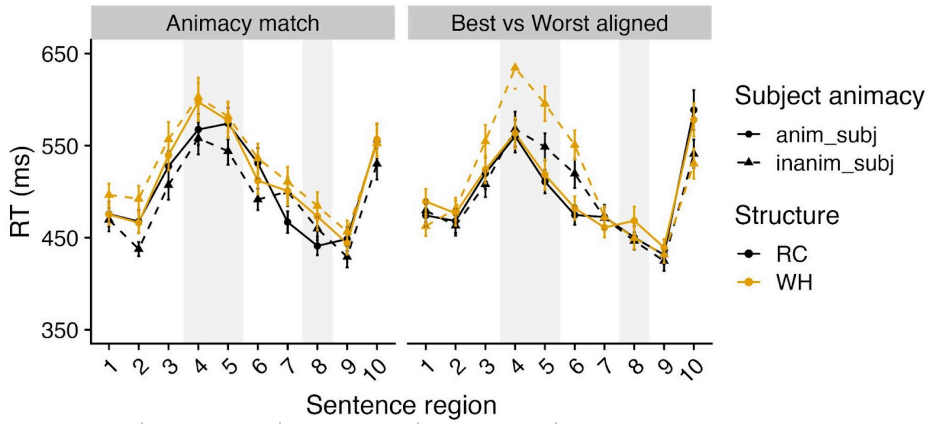


Figure 2 (left). Word-by-word RTs (SEM error bars). Greyed regions are the embedded subject (and its spill-over) and the RP.

Table 2 (below)**.** Exp 1 *brms* output. All other contrasts contain zero.

	Est.	Est. Err	l-95% CI	u-95% CI
structure	-0.31	0.04	-0.38	-0.23
tail	-0.24	0.04	-0.31	0.17
subj_anim	-0.14	0.04	-0.22	-0.07
obj_anim	0.09	0.04	0.02	0.16
stru:tail	-0.18	0.03	-0.23	-0.12
stru:subj	-0.05	0.03	-0.10	-0.00
tail:subj	0.14	0.03	0.08	0.21
stru:obj	0.13	0.02	0.08	0.17
tail:obj	-0.11	0.02	-0.16	-0.07

References:

[1] Pesetsky, D. 1998, *MIT Press.*

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